

Final Report: Delivering Elite European Football Analytics

DSCI 591 Capstone: Trilemma Foundation

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Executive Summary

We built a fully automated football analytics platform for the Trilemma Foundation on top of the StatsBomb open event dataset (StatsBomb 2024). The platform ingests raw match, event, and lineup data from 11 competitions (3,464 matches, approximately 12.2 million on-ball events) into Google BigQuery, transforms it through a three-layer dbt pipeline into analytics-ready tables, enriches those tables with two machine learning models, and serves the results through a Looker Studio dashboard and a plain-English chatbot.

Three research questions guided the project. **RQ1 (Player archetypes)** asked what tactical roles exist across European football and how they differ by position and competition. Unsupervised clustering (PCA + K-Means, $k = 5$) on 13 per-90 features produced five outfield archetypes (Low Activity, Defensive Anchor, Creative Playmaker, Pressing Forward, and Creative Winger) across 5,759 player-seasons, with position as the dominant driver and league context secondary. **RQ2 (Cross-context consistency)** asked whether players maintain similar profiles for club and country. Among 489 dual-context players with ≥ 270 minutes in both settings, 32% rank as Elite in both contexts and 36% as single-context specialists. **RQ3 (Natural language access)** asked whether non-technical analysts could query the dataset without SQL. A Groq-powered Django chatbot returns mart-backed answers in under ten seconds.

The delivered product is reproducible, fully documented, and deployable with a single `docker compose up` command.

Introduction

This section describes the context, motivation, and refined scientific objectives for the project.

Background

The Trilemma Foundation aims to bring data-driven methods to football analysis. StatsBomb open data (StatsBomb 2024) provides one of the richest publicly available sources of match event data,

covering player-level on-ball actions (shots, passes, carries, pressures, dribbles, and more) at sub-second granularity. Despite its depth, the dataset is difficult to work with at scale. Event records are deeply nested JSON, competitions and seasons are fragmented across many files, and extracting meaningful player metrics requires complex aggregation logic that is impractical to write ad hoc.

The Problem

No systematic, production-grade pipeline existed to transform StatsBomb event data into a unified analytical layer accessible to non-technical users. Analysts who could not write SQL against nested schemas were effectively locked out of the data. Trilemma needed a product that could answer tactical questions (which players fit a pressing role, which players perform differently for their country than their club) without requiring custom queries for every request.

Refined Scientific Objectives

Over the course of the project, the initial problem was refined into three concrete and measurable research questions:

1. **RQ1 (Player archetypes):** What tactical archetypes exist across European football, and how does role distribution differ by position and competition?
2. **RQ2 (Cross-context consistency):** Do players maintain similar tactical profiles when playing for their club versus their national team, and which players specialize in one context?
3. **RQ3 (Natural language access):** Can a non-technical analyst query the full dataset in plain English and receive accurate, data-grounded answers in near real time?

These objectives were chosen because they are directly actionable for a football analytics organization. Archetype labelling supports scouting and squad-building decisions, consistency scoring informs national team selection strategy, and the chatbot removes the SQL barrier for day-to-day analytical work.

The data product addresses all three. The Looker Studio dashboard visualizes RQ1 and RQ2 findings interactively, and the chatbot covers RQ3 while also enabling ad-hoc queries against the full mart layer.

Data

This section describes the data source, its structure, quality issues, and how it shaped design decisions.

Key EDA Findings

Coverage is highly uneven across seasons: 2015–16 alone accounts for roughly half of all ingested matches, so EDA and interpretation focus on that cycle where sample depth is greatest ([Figure 1](#)).

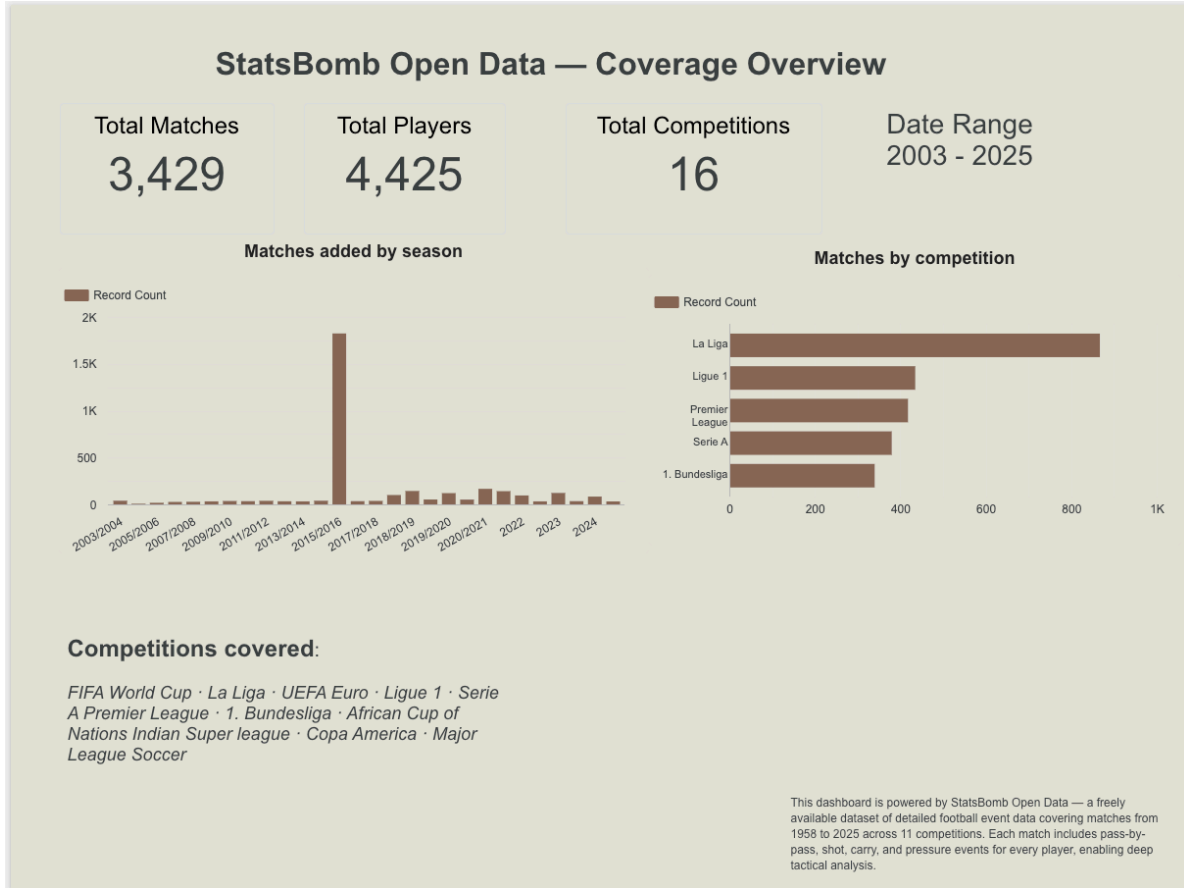


Figure 1: StatsBomb open-data match volume by season and competition. Coverage is uneven, with 2015–16 accounting for roughly half of all matches.

Source

All data in this project comes from the StatsBomb open dataset, accessed via the `statsbombpy` Python library (StatsBomb 2024). Like other public spatio-temporal event repositories (Pappalardo et al. 2019), StatsBomb collects match event data by manually tracking every on-ball action in a match, assigning it a type, timestamp, location (x, y coordinates on a 120×80 pitch grid), and a rich set of type-specific attributes. Our per-90 feature set follows the action-based profiling tradition in football analytics (Decroos et al. 2019). The open tier covers 16 competitions spanning men’s senior football across Europe, South America, North America, and international tournaments.

Dataset Structure

Three tables are ingested from the API (Table 1):

Table 1: StatsBomb tables ingested into BigQuery via `statsbombpy`. The events table drives per-90 feature engineering, matches provide competition context, and lineups supply minutes and position.

Table	Grain	Scale
<code>matches</code>	One row per match	3,464 matches
<code>events</code>	One row per on-ball action	~12.2 million rows
<code>lineups</code>	One row per player appearance	165,820 rows

The `events` table is the analytical backbone. Action types are dominated by Pass (~3.4M), Ball Receipt (~3.2M), Carry (~2.6M), and Pressure (~1.1M). There are 88,023 Shot events with non-null StatsBomb xG values (mean xG per shot: 0.107). The dataset covers 10,808 distinct players across 308 teams. Primary model outputs are **cluster labels** (RQ1), **performance quadrants**, and **consistency scores** (RQ2). Player-level modelling is limited to **men’s senior** competitions, which bounded project scope.

Data Quality and Limitations

Several quality issues and constraints shaped the project:

- **Missing lineup times.** Substitution `to_time` is missing for 110,917 lineup rows, corresponding to starters who played the full match and bench players who never appeared. Minutes played are inferred as 90 (or 120 for extra time) when the field is null.
- **Missing positions.** `position_name` is absent for 34,179 lineup rows, consistent with bench entries with unconfirmed roles. These rows are excluded from per-position analyses.
- **Uneven competition coverage.** La Liga contributes 868 matches with complete seasonal coverage. The UEFA Champions League has scattered single-match history. Analysts should treat early-season or low-coverage competitions as smaller sub-samples.
- **Open data scope.** The free tier does not include all competitions or all seasons available in the paid StatsBomb API. Coverage is curated by StatsBomb and may change with new releases.

How Data Shaped Design Decisions

The minimum 270-minute playing time threshold (equivalent to three full matches) was introduced to address the instability of per-90 rates for players with very few appearances. Below this threshold, a single exceptional match can dominate a player-season’s metrics and produce unrepresentative rate statistics. This threshold is applied consistently across all intermediate dbt models, the clustering pipeline, and the consistency scoring pipeline. The choice of 270 minutes was validated against the distribution of player-season minute totals: it retains the majority of regular contributors while removing statistical noise from small samples.

The nested JSON structure of the `events` table required flattening at ingestion time. Columns containing dictionary objects are normalized using `pd.json_normalize` before writing to Parquet,

and all object-type columns are cast to strings to ensure compatibility with the Apache Arrow schema expected by BigQuery.

Data Science Methods

Our analytical work addresses *how* players play, not just output volume. Two models sit on the same 13 per-90 StatsBomb features (attack, progression, defending, efficiency): **PCA + K-Means** for tactical archetypes and **within-context z-score scoring** for club vs national consistency. Both run on a shared automated pipeline (Figure 2). Full feature definitions, preprocessing steps, and formulas are in the supplementary documentation (Table 6).

Data Pipeline

StatsBomb open data is ingested through GCS into BigQuery, transformed with dbt (staging, intermediate, marts), enriched by Python ML scripts (`cluster.py`, `consistency.py`), and served through a Django app embedding Looker Studio and a Groq chatbot. Dagster orchestrates weekly full runs and hourly refreshes when new files land in GCS. Deployment and maintenance details are in the supplementary documentation (Table 6).

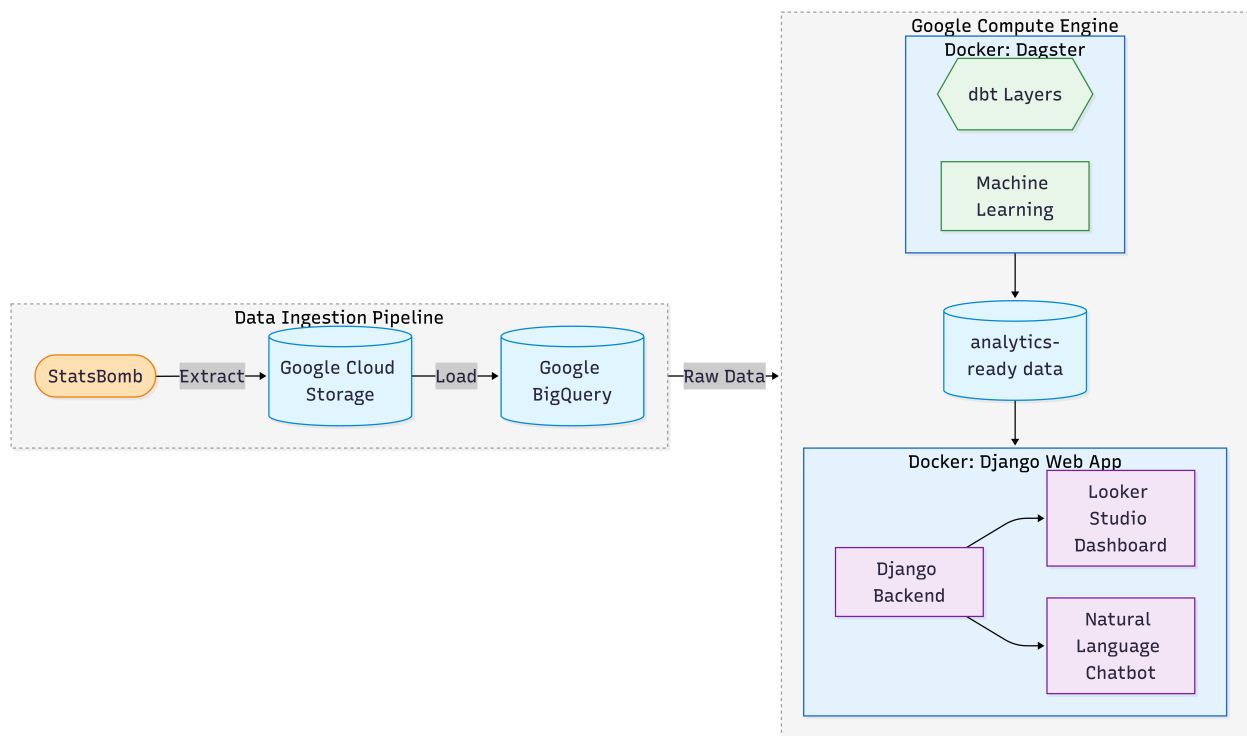


Figure 2: Automated ELT pipeline from StatsBomb open data through BigQuery, dbt, and Python ML scripts to Looker Studio and a Django chatbot, orchestrated by Dagster.

End-to-end validation confirmed aligned row counts (5,759 player-seasons and 489 dual-context players scored). Mart schemas are documented in the supplementary documentation (Table 6).

RQ1: Player Archetype Clustering

This section describes the unsupervised clustering model for RQ1. Findings and dashboard outputs are in [RQ1: Role Distribution](#).

We cluster player-seasons from `int_player_season_stats` after applying a **270-minute** minimum per season ($n = 5,469$ outfield, 5,759 including goalkeepers). **Goalkeepers** are excluded from K-Means and assigned a fixed label because their event profiles are structurally unlike outfield players.

Feature selection. Thirteen per-90 metrics span attacking output (`xg_per_90`, `shots_per_90`), ball progression (`passes_per_90`, `passes_att_third_per_90`, `pass_completion_pct`, `carries_per_90`, `dribbles_per_90`), defensive work (`pressures_per_90`, `interceptions_per_90`, `blocks_per_90`, `clearances_per_90`, `duels_per_90`), and shot efficiency (`xg_per_shot`). The same 13 features are used in clustering, consistency scoring, and the dashboard so labels stay aligned across the product.

Preprocessing and modelling: missing values are median-imputed to preserve the zero-skewed shape of defensive counts. Values above the 99th percentile are clipped column-wise to limit outlier influence, then features are standardised. PCA (Jolliffe and Cadima 2016) retains components explaining 80% of variance (typically six). **PC1** captures overall involvement volume. **PC2** contrasts defensive actions against attacking output. K-Means (Lloyd 1982) with $k = 5$ is applied to the PCA-reduced space. We compared $k = 3-7$ using silhouette score; $k = 5$ offered the best trade-off between cluster separation and coach-facing interpretability ([Notebook 05](#)). K-Means was preferred over hierarchical methods for scale and reproducibility on 5,000+ player-seasons.

Table 2: RQ1 archetype distribution across 5,759 player-seasons. Outfield players receive K-Means labels and goalkeepers a fixed label. Low Activity is the largest archetype and Creative Winger the smallest outfield group.

Archetype	Seasons	Tactical profile
Low Activity	1,963	Low volume across all metrics. Typical of squad players and limited roles
Defensive Anchor	1,470	High clearances, interceptions, duels. Low attacking output
Creative Playmaker	807	High pass completion, progressive passes, carries. Attacking midfielders
Pressing Forward	766	High pressures, shots, and xG. High-energy forwards and pressing wingers
Creative Winger	463	High dribbles, carries, shots. Direct wide attackers
Goalkeeper	290	Assigned by position. Excluded from K-Means

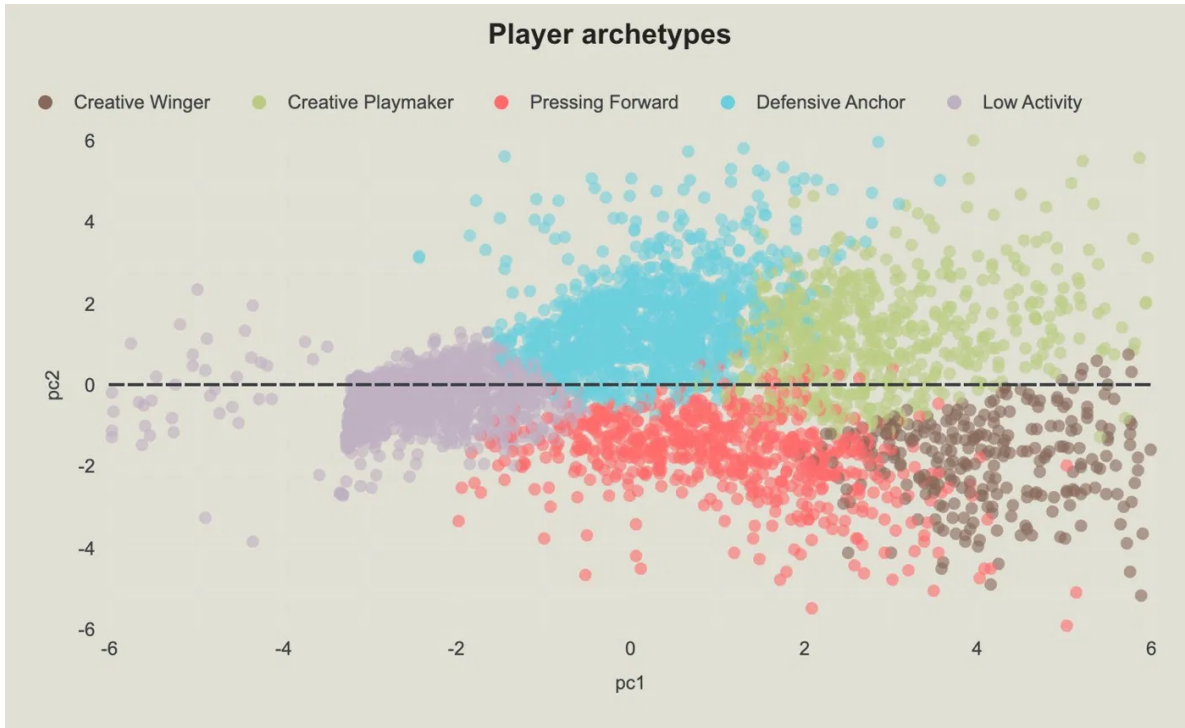


Figure 3: Looker cluster explorer showing outfield player-seasons in PCA space (PC1 vs PC2), coloured by K-Means archetype label ($k = 5$).

The resulting archetypes (Table 2) feed the Looker cluster explorer and chatbot marts. Archetypes describe *how a player is used in a given season*, not a fixed career identity. Lionel Messi is mostly classified as Creative Winger but as Pressing Forward during Copa América 2024, consistent with Argentina’s more compact national-team system in that tournament.

RQ2: Club vs National Consistency Scoring

This section describes the within-context scoring model for RQ2. Findings and dashboard outputs are in [RQ2: Club vs National Consistency](#).

For **489 dual-context players** with ≥ 270 minutes in both club and national football, we standardise each of the 13 features **within context**. Club z-scores compare a player only to other club players. National z-scores compare only to national-team peers.

Z-score standardisation (within each context):

$$Z = \frac{x - \mu_{\text{context}}}{\sigma_{\text{context}}}$$

Performance score (club and national axes on the scatter plot):

$$\text{performance score} = \sum_{i=1}^{13} (Z_i \times w_i)$$

where Z_i is the z-score for per-90 metric i and w_i is its PCA-derived weight (same 13 metrics as RQ1 clustering).

Consistency score (profile similarity, not quality level):

$$\text{consistency score} = 1 - \frac{1}{13} \sum_{i=1}^{13} |Z_{i,\text{club}} - Z_{i,\text{national}}|$$

Quadrants split the cohort at the **medians** of club and national performance score ([Table 3](#)):

Table 3: Quadrant rules from `src/ml/consistency.py`. Dual-context players are classified by whether club and national performance scores fall above or below cohort medians on the same 13 per-90 metrics.

Club vs median	National vs median	Quadrant
$\geq$ median	$\geq$ median	Elite
$\geq$ median	$<$ median	Club Specialist
$<$ median	$\geq$ median	International Specialist
$<$ median	$<$ median	Underperformer

Outputs were sanity-checked against known dual-context players (e.g., Gnabry as International Specialist). Progressive passing into the attacking third best separates club-focused from national-focused specialists in exploratory comparisons. Quadrant labels reflect peer standing within each context, not absolute quality. An Elite player outperforms cohort medians in both settings without any claim about raw output level. The 270-minute threshold removes rotation players, stabilising

per-90 estimates at the cost of excluding low-minute prospects relevant to scouting work. Complete formulas and dashboard field definitions are in the supplementary documentation ([Table 6](#)).

Stakeholders and ethics. Trilemma analysts and partner scouts use archetypes for scouting decisions, and national-team staff use consistency scores for selection. Players are indirectly affected by labels, so outputs are scoped to playing style, not player value. The project is limited to men’s senior competitions per StatsBomb open-data coverage and terms of use.

Data Product and Results

The data product is a unified Django web application combining an embedded Looker Studio dashboard and a natural language chatbot. These forms were chosen because Trilemma’s analysts operate at two levels of inquiry that no single tool handles alone: broad visual exploration and specific ad-hoc questions. A notebooks-only approach was ruled out as inaccessible to non-technical partner staff. A dashboard alone could not replace flexible querying without Looker edit access.

Results

RQ1: Role Distribution

See also [RQ1: Player Archetype Clustering](#). Chi-square tests and Cramér’s V on top-five European leagues show **position drives archetype assignment** more strongly than competition (full heatmaps and statistics in [Notebook 07](#)). In row-normalised heatmaps:

- **Center backs** concentrate in Defensive Anchor and Low Activity. Attacking clusters are rare.
- **Forwards and attacking mids** skew toward Pressing Forward and Creative Winger.
- **Central mids** spread across Creative Playmaker, Pressing Forward, and Low Activity, reflecting overlapping box-to-box roles.
- **Full backs** often land in Creative Winger or Creative Playmaker when their profiles emphasise carries and progressive passing.

League effects are secondary but visible ([Table 4](#)). Serie A has the largest Low Activity share among the top five. Bundesliga and La Liga show relatively more Pressing Forwards. The same position can shift ± 5 –15 percentage points in archetype share between leagues. Dashboard users should filter by position before comparing leagues.

Table 4: Outfield archetype counts by top-five European league (goalkeepers excluded). Defensive Anchor and Low Activity lead in every league, with Serie A showing the highest Low Activity share.

Competition	Largest (seasons)	Second largest	Notable pattern
La Liga	Defensive Anchor (304)	Low Activity (167)	Highest volume. Defensive Anchors lead
Serie A	Low Activity (240)	Defensive Anchor (138)	Most Low Activity share among top five
Ligue 1	Defensive Anchor (175)	Low Activity (148)	Near-even top two clusters
Premier League	Defensive Anchor (169)	Low Activity (163)	Near-even top two clusters
Bundesliga	Defensive Anchor (145)	Low Activity (121)	Smallest top-five sample. More Pressing Forward (83)

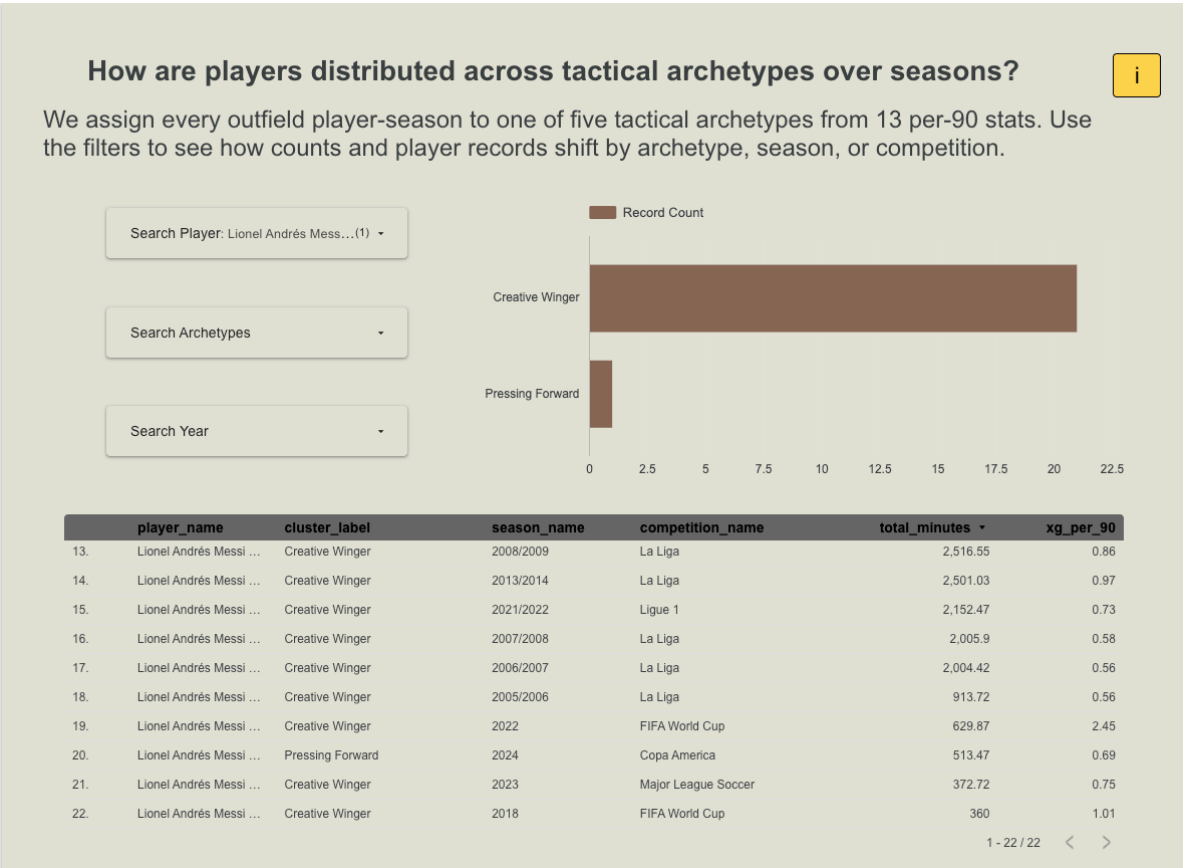


Figure 4: Messi player-seasons in the cluster explorer. Most are Creative Winger, but Copa América 2024 is Pressing Forward, showing archetypes reflect seasonal role rather than fixed identity.

RQ2: Club vs National Consistency

See also [RQ2: Club vs National Consistency Scoring](#). Of 489 dual-context players, **32% are Elite** (above peer median in both contexts), **32% Underperformer**, and **36% single-context specialists** (18% club-only, 18% national-only, see [Table 5](#)). About two thirds of the cohort rank

above or below peer median in **both** settings rather than in only one. Comparing Club Specialist vs International Specialist players, the largest gaps are in **passes into the attacking third per 90**, **interceptions per 90**, and **carries into the final third per 90**. System and role differences, not raw goal output, drive the split.

A stable profile is not the same as a strong one. Many players look similar across contexts while ranking below average in both. Defenders show slightly higher mean consistency than forwards (Notebook 08), suggesting attacking roles shift more between club tactics and national setups. The dashboard’s consistency score and context-shift index measure different things (see supplementary documentation, Table 6).

Table 5: RQ2 quadrant counts for 489 dual-context players with at least 270 minutes in both settings. Elite and Underperformer each account for about one third of the cohort.

Quadrant	Players	Share
Elite	157	32.1%
Underperformer	156	31.9%
Club Specialist	88	18.0%
International Specialist	88	18.0%

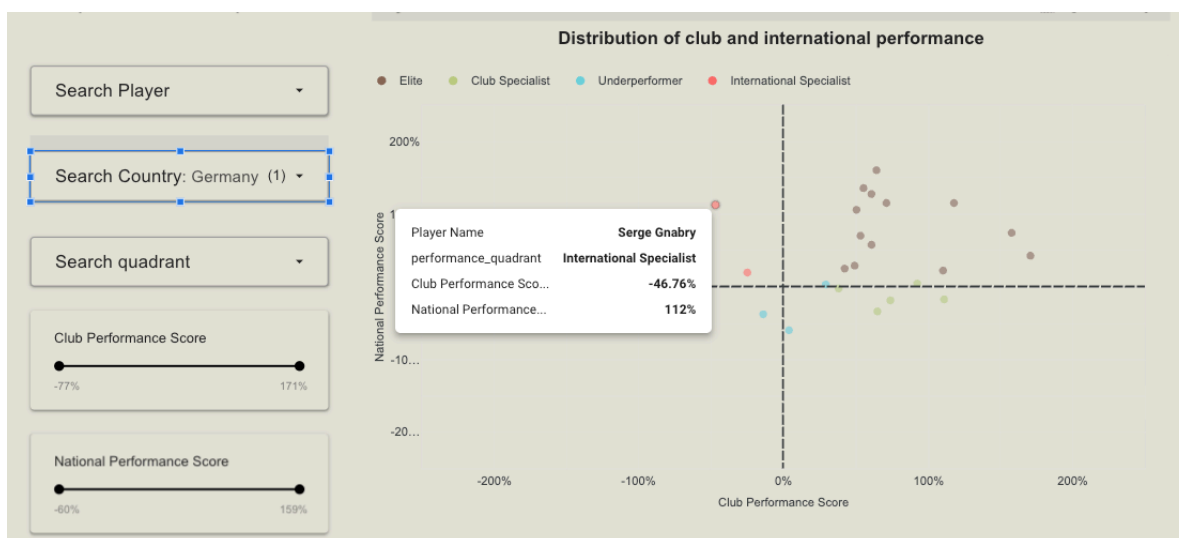


Figure 5: Consistency Explorer for Germany dual-context players. Serge Gnabry appears as an International Specialist, stronger vs national-team peers than club peers.

RQ3: Natural Language Access

The chatbot returns BigQuery-backed answers to plain-English questions in under ten seconds. It was tested manually on representative analyst questions (player rankings by archetype, cross-context comparisons, competition filters) and consistently applied the same eligibility rules as the

dashboard, including the 270-minute threshold. Advanced Mode exposes generated SQL for audit. A read-only SQL guard blocks destructive queries.

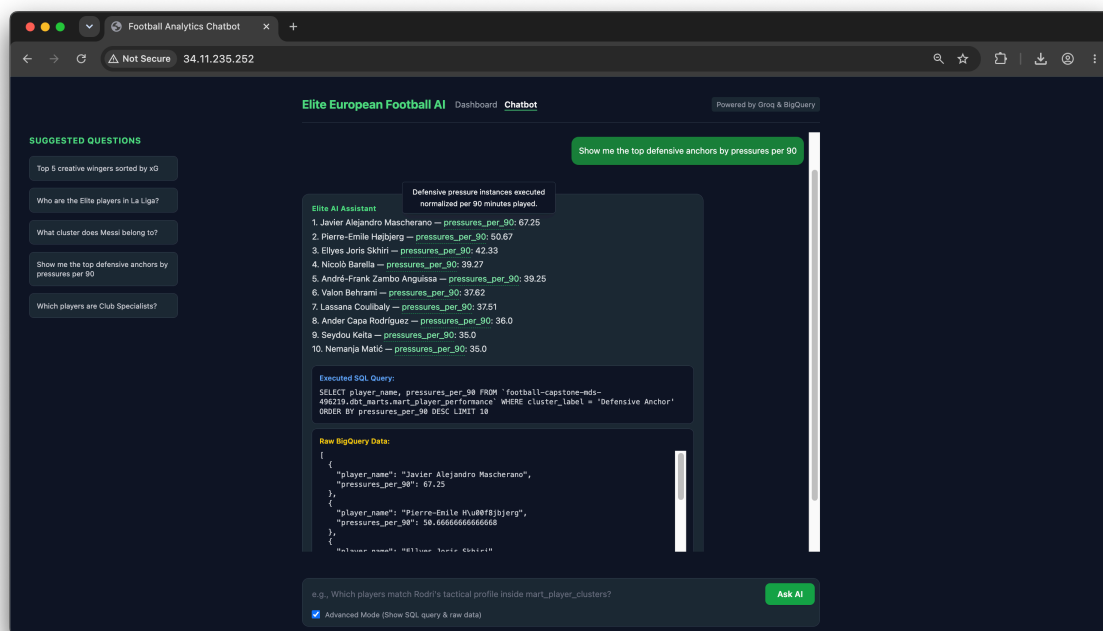


Figure 6: RQ3 chatbot with Advanced Mode showing a natural-language query, formatted top-10 results, and the generated BigQuery SQL for audit.

Product Design

The **dashboard** provides visual access to pre-computed results (archetype distributions, consistency quadrants, and player rankings) filterable by position, competition, and archetype without writing code. It is appropriate for regular reporting and for communicating findings to non-technical stakeholders within Trilemma's partner organizations.

The **chatbot** covers ad-hoc questions outside fixed dashboard charts, returning mart-backed answers with SQL audit and read-only guards for trust.

Strengths: the Dagster-automated pipeline refreshes all data weekly with zero manual intervention, so both interfaces always reflect current StatsBomb releases. The product is fully reproducible from a single `docker compose up` and documented for independent use by a new team member.

Limitations: the chatbot handles single-step queries only — multi-turn reasoning would require session state and prompt chaining beyond the deployment timeline. Both interfaces are scoped to men's competitions; a women's pipeline needs a separate dbt source filter and re-running all intermediate models. Pre-defined dashboard charts cannot be customised without Looker Studio edit access, which was not provisioned for this phase.

Conclusions and Recommendations

The original problem was that Trilemma Foundation analysts had no scalable way to extract tactical insights from StatsBomb event data without writing complex SQL. This project addressed that problem by delivering three research products on a single automated pipeline.

RQ1 established that five tactical archetypes emerge from 5,759 player-seasons across 11 competitions, with position being the dominant driver of cluster assignment. Trilemma can now use archetype labels to profile scouting targets and assess squad balance before a transfer window, replacing ad-hoc SQL with a filterable Looker dashboard. **RQ2** found that 36% of dual-context players specialize in one setting (club or national) while 32% perform above peer median in both, providing a data-grounded tool for national team selection decisions. **RQ3** demonstrated that a Groq LLM reliably translates plain English questions into accurate BigQuery SQL, removing the technical barrier for non-technical analysts entirely.

The product addresses the core needs described in the project brief. Remaining limitations are men’s-only data scope, uneven StatsBomb open-tier coverage, and the chatbot’s single-step reasoning constraint. The models address analytical questions within a defined scope: archetypes profile playing style, not transfer value or injury risk; consistency scores inform national-team context, not squad chemistry or fitness. These boundaries are inherent to the available data, not failures of the models themselves.

Future work should prioritize a women’s parallel dbt pipeline branch, multi-turn LLM reasoning with session memory for follow-up questions, integration with the paid StatsBomb API for real-time data, and extending the consistency framework to track how individual archetypes evolve across seasons. A key lesson from the project is that product form matters as much as model quality. Combining a visual dashboard with a conversational interface gives analysts coverage that neither tool provides independently.

References

- Decroos, Tom, Lotte Bransen, Jan Van Haaren, and Jesse Davis. 2019. “Actions Speak Louder Than Goals: Valuing Player Actions in Football.” In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 1851–61. <https://doi.org/10.1145/3292500.3330758>.
- Jolliffe, Ian T., and Jorge Cadima. 2016. “Principal Component Analysis: A Review and Recent Developments.” *Philosophical Transactions of the Royal Society A* 374 (2065): 20150202. <https://doi.org/10.1098/rsta.2015.0202>.
- Lloyd, Stuart P. 1982. “Least Squares Quantization in PCM.” *IEEE Transactions on Information Theory* 28 (2): 129–37. <https://doi.org/10.1109/TIT.1982.1056489>.
- Pappalardo, Luca, Paolo Cintia, Alessio Rossi, Emanuele Massucco, Paolo Ferragina, Dino Pedreschi, and Fosca Giannotti. 2019. “A Public Data Set of Spatio-Temporal Match Events in Soccer Competitions.” *Scientific Data* 6: 236. <https://doi.org/10.1038/s41597-019-0247-7>.

StatsBomb. 2024. “StatsBomb Open Data.” <https://github.com/statsbomb/open-data>.

Appendix: Supplementary Documentation

Detailed formulas, pipeline architecture, full RQ1/RQ2 analysis notebooks, and mart schemas live in the repository. Links are collected in [Table 6](#).

Table 6: Repository documentation index with links to detailed supplementary materials.

Topic	Document
Consistency score, quadrants, and z-scores	consistency-explorer.md
Context shift score	context-shift-score.md
Consistency pipeline (<code>consistency.py</code>)	docs/consistency.md
RQ1/RQ2 notebooks and exports	docs/notebooks.md
ELT architecture and deployment	HANDOVER.md
Mart table schemas	docs/marts_data_dict.md